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DATS 4001

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Comparative Analysis of Modeling Methods for Local House Price Index Forecasting

1. Objective

This capstone project focused on comparing the accuracy of different time series and machine learning models for forecasting the Washington-Arlington-Alexandria Metropolitan Statistical Area (MSA) housing market, using the All-Transactions House Price Index (House Index / ZHVI) as the target variable. The primary goals were twofold: first, to identify the best-performing model based on walk-forward validation using Relative Mean Squared Error (RelMSE) against a historical mean benchmark over an eight-quarter horizon, and second, to generate an 8-quarter forecast through Q4 2026 using that optimal model. The project initially began with a broader aim of data mining the local housing market but evolved to concentrate on comparing forecasting methodologies after encountering challenges in the preliminary analysis phase.

2. Methodology

2.1 Data Collection & Preprocessing

The foundation of this analysis was quarterly data collected from 1990 Q1 to 2024 Q4. The target variable, the House Index for the specified MSA, was supplemented by additional economic indicators. These included local variables such as Unemployment Rate, CPI, Median Household Income, and Population (sourced from BLS and Census Bureau), along with federal indicators like GDP, Interest Rate, Mortgage Rate, and Poverty Rate (sourced from BEA, FRED, Freddie Mac, and Census Bureau).

A significant challenge arose from combining data originating from different sources and reported at various frequencies (annual, monthly, weekly, quarterly). This inevitably led to missing values, particularly for annually reported data like Median Household Income and Poverty Rate, which hadn't been updated past 2023 at the time of analysis. To create a unified quarterly dataset suitable for modeling, several preprocessing steps were necessary.

All data was first resampled to a Quarterly Start ('QS') frequency. Linear interpolation served as the primary method to fill gaps within the historical data range after resampling, effectively drawing a straight line between known data points to estimate intermediate values. For values missing at the very end of the series (tail-end missingness), especially for the annually reported data, AutoARIMA models were fitted to the available historical data. These models were then used to forecast and fill these recent gaps, extending the dataset forward to ensure completeness for modeling and subsequent multi-step forecasting.

Specific handling varied by variable:

- House Index: Already quarterly, but needed resampling to standardize to specific dates (January, April, July, October) as some years had irregular reporting dates. Linear interpolation filled any resulting gaps.

- CPI: Monthly data resampled to quarterly with linear interpolation.
- Poverty Rate: Annual data resampled to quarterly with linear interpolation for historical gaps and AutoARIMA fill for recent gaps (data ended in 2023).
- Median Household Income: Annual data resampled to quarterly with linear interpolation for historical gaps and AutoARIMA fill for recent gaps (data ended in 2023).
- Population: Quarterly data with linear interpolation for most gaps. For the 1990-2000 period where data was extremely sparse, pure linear interpolation was used to span this 10-year gap.
- Interest Rate: Monthly data resampled to quarterly with linear interpolation.
- Mortgage Rate: Weekly data first averaged to monthly means, then merged to quarterly frequency.
- GDP: Quarterly data with linear interpolation for minor historical gaps.
- Unemployment Rate: Monthly data resampled to quarterly with linear interpolation.

In hindsight, while AutoARIMA filling was more sophisticated than simple forward filling, there might have been better approaches for imputing certain variables like unemployment rate, mortgage rate, or interest rate, which have their own unique patterns that AutoARIMA might not fully capture.

2.2 Stationarity, Transformation, and Feature Engineering

Ensuring data stationarity—meaning the statistical properties like mean and variance remain constant over time—was crucial for time series models like ARIMA, VECM, and VAR. Initial analysis using Augmented Dickey-Fuller (ADF), KPSS tests, and insights from AutoARIMA indicated that most variables were non-stationary in their original form. The Unemployment Rate

was the exception, found to be stationary ($I(0)$). Most other variables required first differencing ($d=1$), while GDP required second differencing ($d=2$).

However, initial attempts with VECM/VAR models using this differenced data still resulted in problematic residuals, suggesting underlying issues like structural breaks or non-linearities weren't fully addressed. This led to a strategic pivot towards using percentage changes for most variables (excluding the stationary Unemployment Rate) when preparing data for the regression and machine learning models. Stationarity tests largely confirmed the suitability of these percentage-change series.

For the machine learning and regression models (Lasso, ElasticNet, SVR, LSTM), further feature engineering was performed. Lags 1 through 8 were created for the target variable (House Index % change) and all exogenous predictors to capture historical information spanning up to two years. A simple integer time trend was also added. Additionally, recognizing the potential impact of major economic disruptions, dummy variables were introduced to account for structural breaks during the 2008 financial crisis (2007 Q4 - 2009 Q4) and the COVID-19 pandemic (2020 Q1 - 2021 Q4).

2.3 Modeling Approaches and Evaluation Framework

A diverse set of modeling approaches was evaluated using a rigorous walk-forward validation framework to simulate real-world forecasting performance. The data was initially split into a training set (1990 Q1 - 2015 Q4) and a test set (2016 Q1 - 2024 Q4). Models were trained on the initial set, generated an 8-step forecast, then retrained including one more quarter, forecasted again, and so on, iterating through the test set. This process used nine folds based on Time Series Split principles to respect the temporal order of the data.

A historical mean forecast served as the benchmark. This simple model calculated the average percentage change from the available training data at each step and projected this average forward iteratively for 8 quarters.

Model performance was primarily evaluated using Relative Mean Squared Error (RelMSE), calculated as the model's MSE divided by the benchmark's MSE for each forecast step (1 through 8). A RelMSE below 1 signifies that the model outperformed the benchmark. Other metrics like RMSE, MAE, MAPE, and Directional Accuracy were also examined.

The models compared included:

AutoARIMA: Used for baseline comparison and imputation.

VECM/VAR: Explored for econometric system modeling. VECM (Vector Error Correction Model) is a complex econometric model that, in the simplest terms, uses vectors to correct for errors in the system. It's designed to capture the dynamic relationships between multiple time series variables that are cointegrated. VAR (Vector Autoregression) is a related approach that models multiple time series variables and their lagged values simultaneously.

Lasso & ElasticNet Regression: Linear models with regularization for feature selection and handling multicollinearity. Lasso uses L1 regularization, which can reduce variable coefficients to exactly zero, effectively performing feature selection. ElasticNet combines L1 (Lasso) and L2 (Ridge) regularization, making it particularly useful when dealing with multicollinear variables.

Hyperparameters were tuned using RandomizedSearchCV within each walk-forward step.

Support Vector Regression (SVR): A machine learning model adept at capturing non-linear relationships, specifically configured to use only lags 1-3 of the target variable based on PACF analysis. Hyperparameters were tuned once before the validation loop.

Long Short-Term Memory (LSTM): A deep learning model designed for sequential data that can capture long-term dependencies, tuned using Bayesian optimization within each validation step.

3. Results and Findings

3.1 Model Performance Walk-Forward

The walk-forward validation revealed distinct performance characteristics for each model class.

ARIMA: AutoARIMA provided a solid baseline, automatically identifying appropriate model orders (e.g., $(0,1,3)(1,0,0)[4]$ for House Index). Its forecasting performance was reasonable, particularly strong at the 2-step horizon in this run, but generally did not match the top machine learning models.

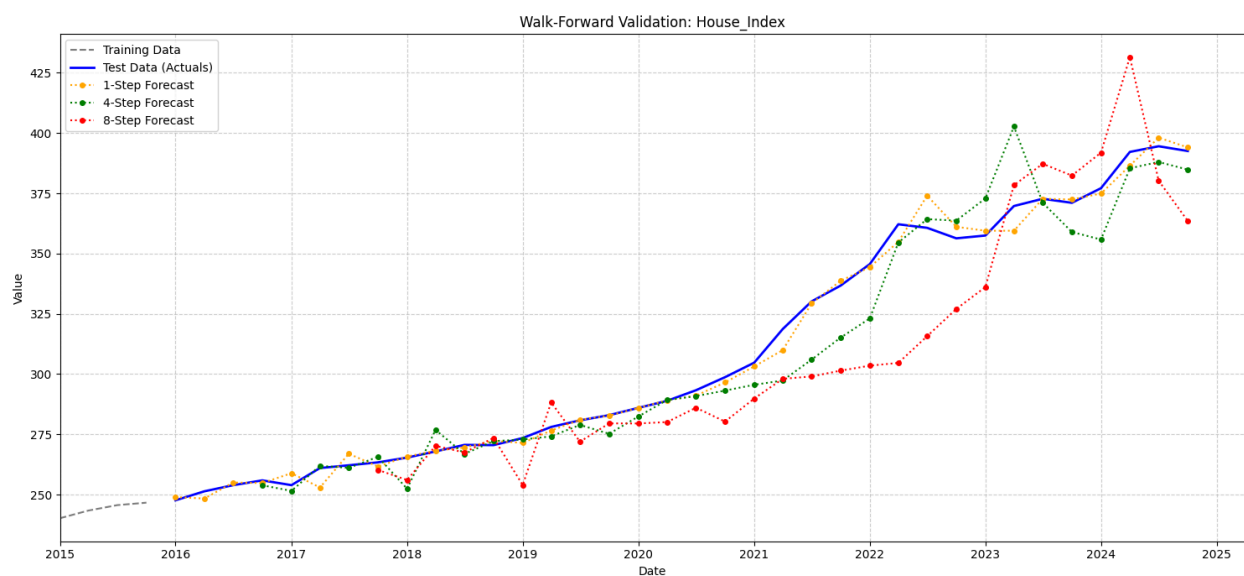


Figure 1: ARIMA's Walk-Forward Validation for House Index

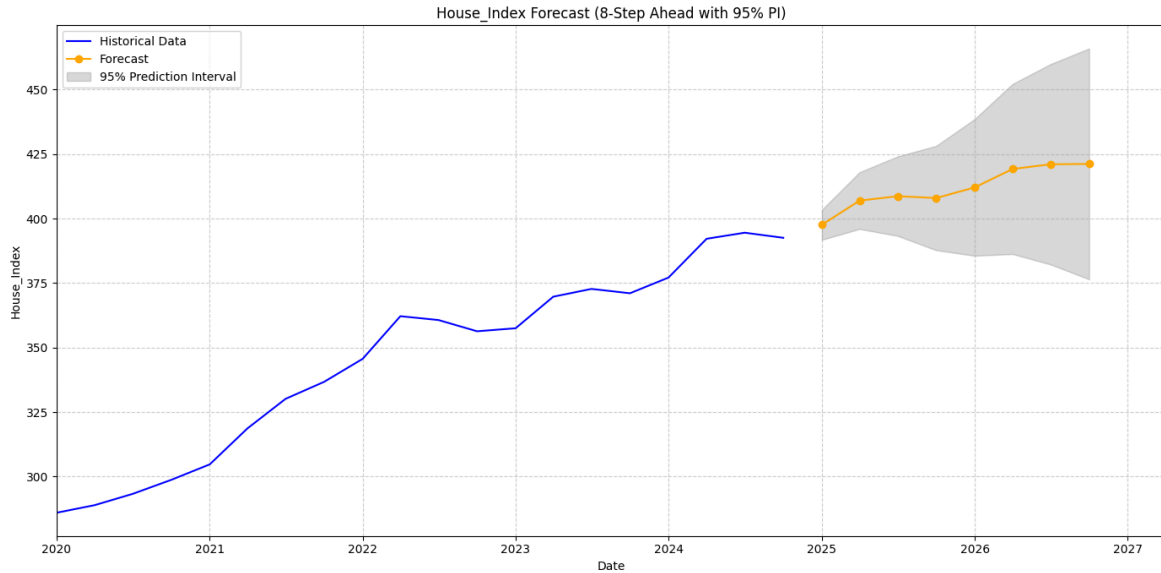


Figure 2: Final ARIMA Forecast for House Index

VECM/VAR: These complex econometric models proved unsuitable for this dataset. Despite exploring differenced data and percentage changes, the model residuals consistently failed diagnostic tests for normality and autocorrelation. The presence of patterns and potential structural breaks in the residuals indicated model misspecification, rendering them unreliable for forecasting. These models are extremely sensitive to their underlying assumptions, and despite my economics background, implementing them properly was quite challenging.

VECM Residuals Over Time (with Exogenous Variables)

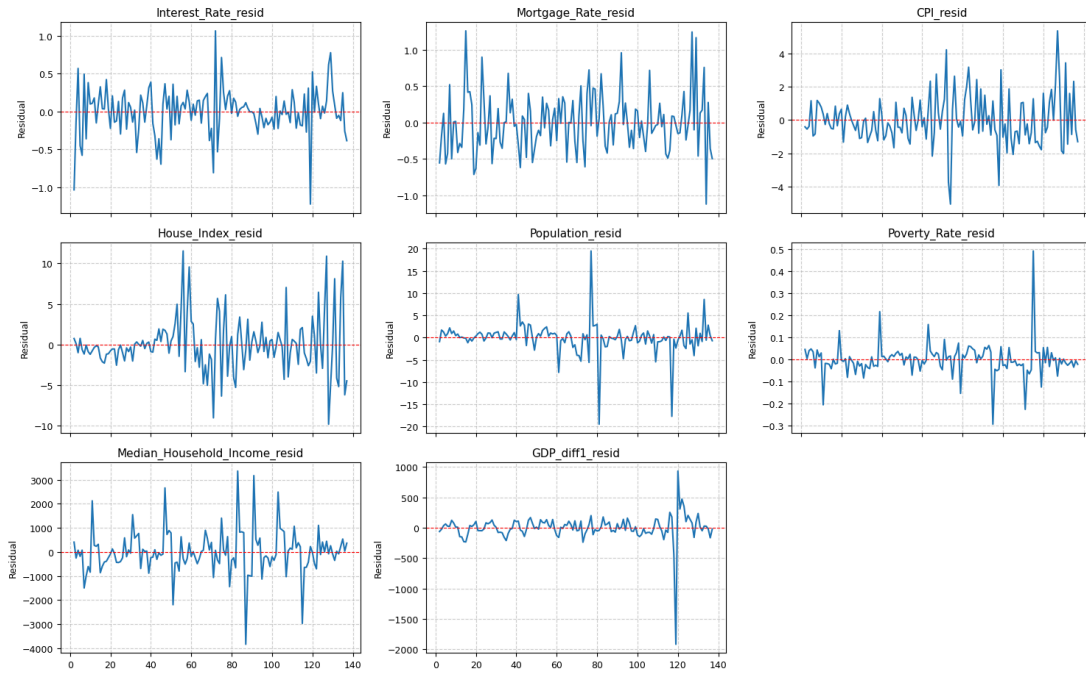


Figure 3: VECM Model Residuals, after differencing

VAR Model Residuals

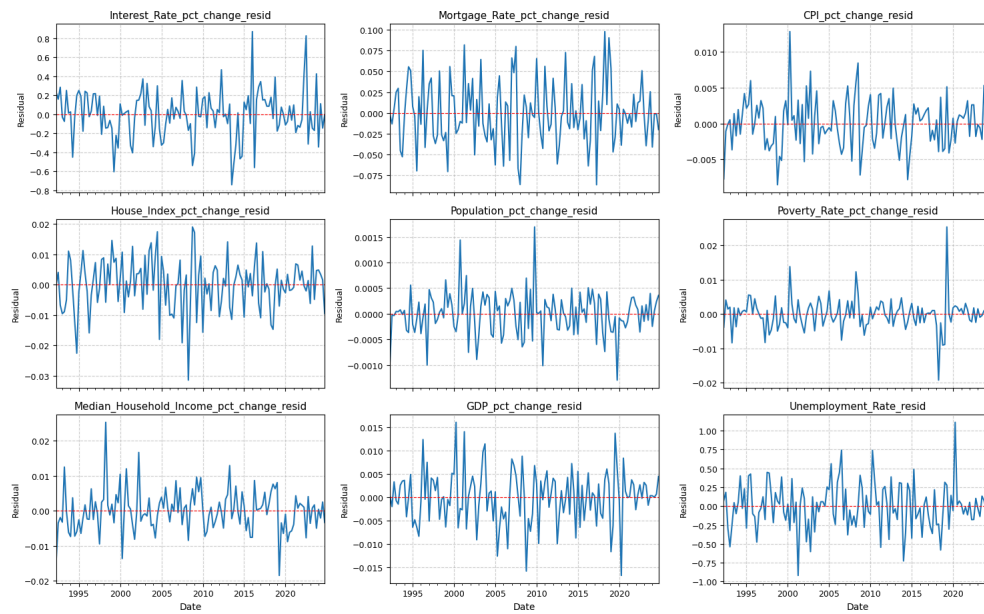


Figure 4: VAR Model Residuals, after percent change differencing

ACF/PACF Insights & Feature Selection: Analysis of the House Index percentage change using Autocorrelation (ACF) and Partial Autocorrelation (PACF) plots was informative and became a critical turning point in the modeling strategy. The ACF plot showed significant correlation extending to lag 12, suggesting strong temporal dependencies. However, the PACF plot, which isolates the direct influence of a lag by removing effects from intermediate lags, clearly showed significance only up to lag 3.

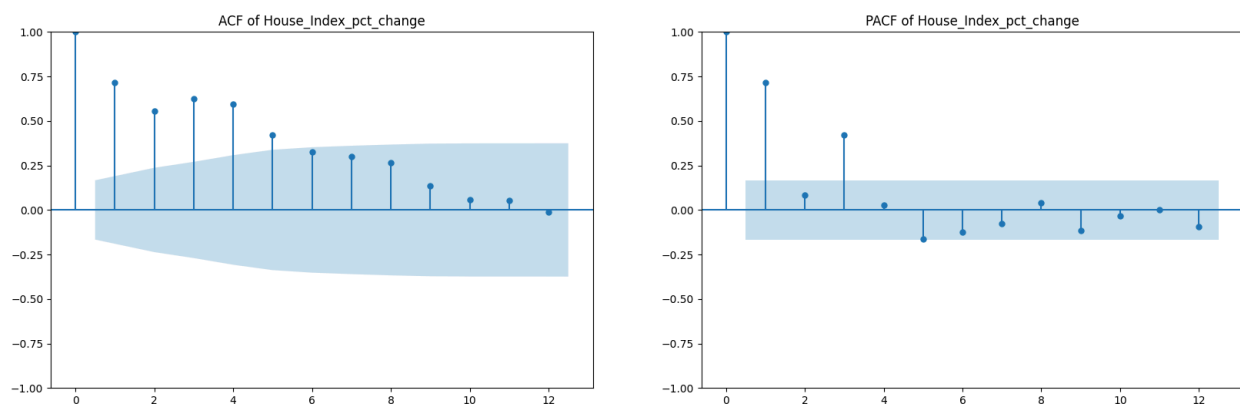


Figure 5: ACF and PACF plot showing autocorrelation on percent change of House Index

This PACF finding proved crucial for the subsequent modeling approaches, particularly for the SVR implementation. Independently, the Lasso and ElasticNet models, through their automated L1 regularization feature selection process, strongly favored lags 1, 3, and sometimes 4 of the House Index percentage change, largely ignoring other economic indicators. This convergence validated the PACF findings and underscored the strong autoregressive nature of the target variable.

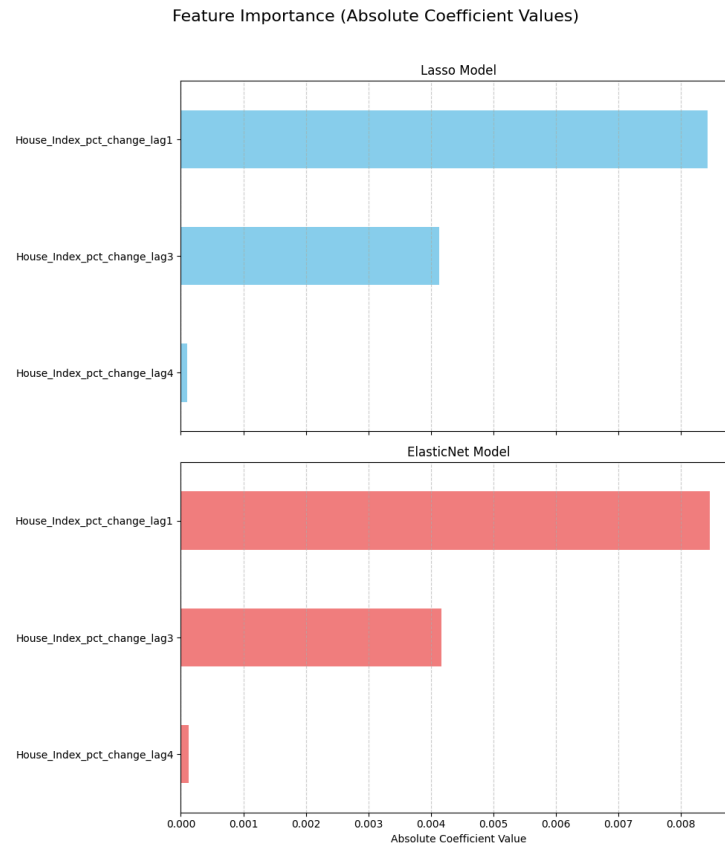


Figure 6: Feature Importance Ranked by Lasso and ElasticNet Regression

SVR: Guided by the PACF analysis, the SVR model was trained using only the first three lags of the House Index percentage change. Tuned only once before the validation loop, it delivered surprisingly consistent and robust performance, achieving the best RelMSE relative to the benchmark across most forecast horizons. The simplicity of using just three autoregressive terms, combined with SVR's ability to capture non-linear relationships, proved to be a powerful combination for this specific forecasting task.

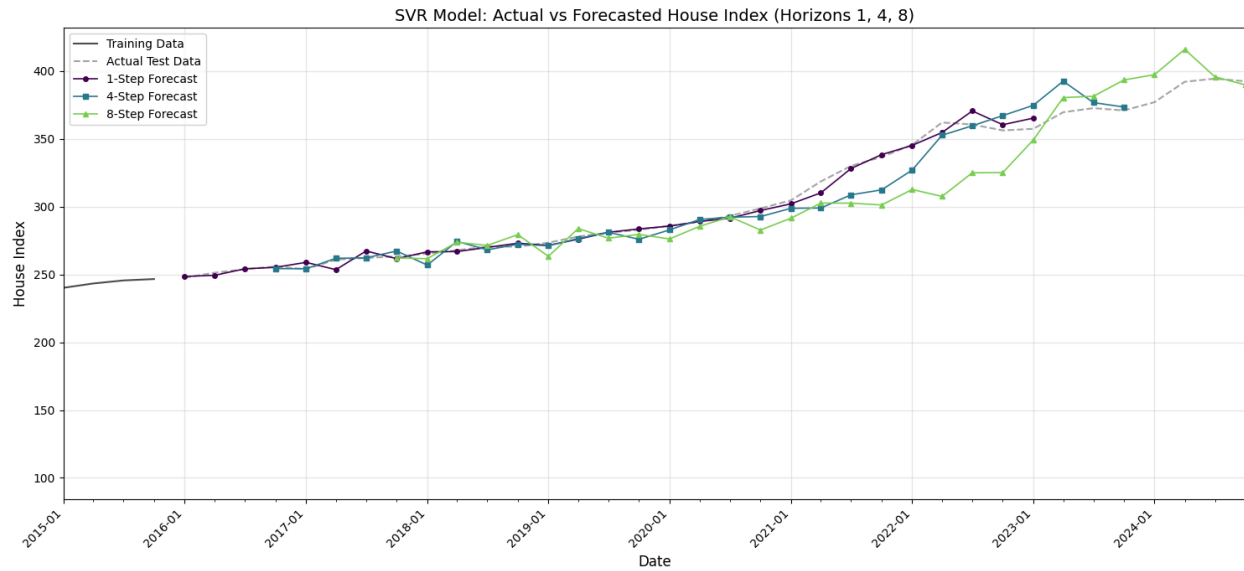


Figure 7: Walk-Forward Validation Results on SVR

LSTM: The deep learning approach required computationally intensive step-wise hyperparameter tuning. While ranking second at the 1-step horizon, its performance degraded substantially at longer horizons, placing it last overall among the evaluated models in this implementation. This may reflect challenges in tuning the model appropriately or the possibility that the housing price data doesn't exhibit the kind of long-term dependencies that LSTM is designed to capture.

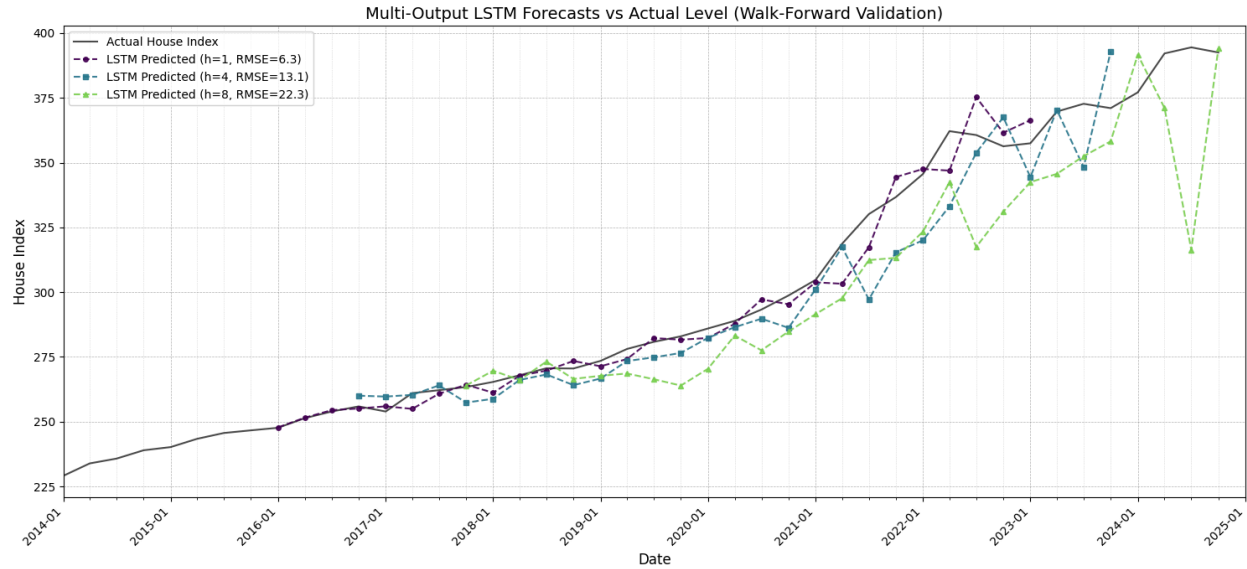


Figure 8: Walk-Forward Validation Results on Steps 1,4 and 8 on LSTM

3.2 Final Model Selection and Forecast

Given its superior and consistent performance in the walk-forward validation, Support Vector Regression (SVR) using lags 1-3 of the House Index % change was selected as the final, best-performing model.

Retraining this SVR model on the full dataset up to 2024 Q4 yielded an 8-quarter forecast projecting continued, albeit potentially moderating, growth. The House Index is forecasted to reach approximately 424.88 by Q4 2026.

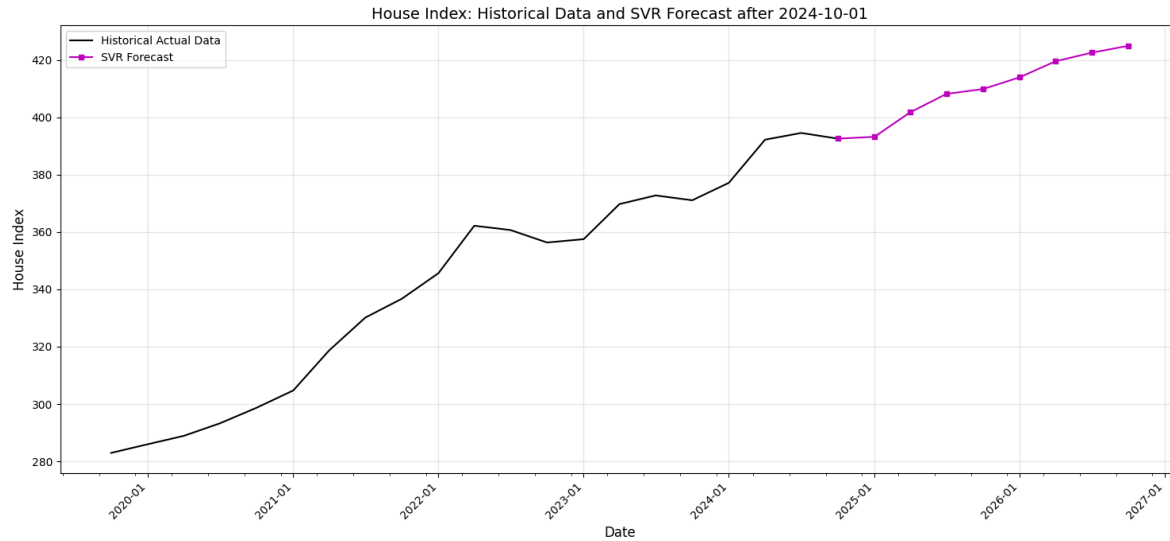


Figure 9: Final Forecast Using Highest Performing Model (SVR)

Table 1: Final SVR Forecast Values (Rounded)

Date	SVR Forecast
2025-01-01	393.17
2025-04-01	401.74
2025-07-01	408.14
2025-10-01	409.82
2026-01-01	413.92
2026-04-01	419.50
2026-07-01	422.51
2026-10-01	424.88

From a practical perspective, this forecast suggests that the DC-area housing market will continue its upward trajectory through 2026. For potential homebuyers, this implies that waiting for significant price drops may not be a viable strategy. However, the forecast also suggests some seasonal patterns, with prices tending to stagnate slightly in the second half of the year, which could offer slightly better buying opportunities during those periods.

3.3 Overall Model Ranking

The Relative MSE ranking table summarizes the comparative performance. SVR's dominance, especially for steps 1 through 6, is evident. Most sophisticated models added value over the simple historical mean initially, but this advantage diminished as the forecast horizon extended. At the 8-step horizon, SVR's accuracy was essentially identical to the benchmark, highlighting the difficulty in outperforming a simple trend extrapolation far into the future for a series with such strong historical upward momentum, despite short-term fluctuations.

Table 2: Model Performance Ranking (Relative MSE) Across Prediction Steps

Steps	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
1	SVR (0.791)	LSTM (0.874)	ElasticNet (0.886)	ARIMA (0.907)	Lasso (0.923)
2	SVR (0.676)	ARIMA (0.948)	ElasticNet (0.975)	Lasso (1.000)	LSTM (1.074)
3	SVR (0.768)	ElasticNet (0.955)	Lasso (0.970)	ARIMA (0.990)	LSTM (1.209)
4	SVR (0.774)	ElasticNet (0.867)	Lasso (0.873)	ARIMA (1.028)	LSTM (1.038)
5	SVR (0.858)	ElasticNet (0.994)	Lasso (1.007)	LSTM (1.031)	ARIMA (1.259)
6	SVR (0.966)	ElasticNet (1.095)	Lasso (1.112)	LSTM (1.166)	ARIMA (1.429)
7	SVR (1.013)	Lasso (1.093)	ElasticNet (1.097)	LSTM (1.450)	ARIMA (1.513)
8	SVR (1.000)	Lasso (1.087)	ElasticNet (1.092)	ARIMA (1.409)	LSTM (1.493)

Note: Relative MSE (shown parenthetically in Table 2) is calculated against a historical means benchmark model. This benchmark uses the average percentage change from historical data, is multi-outputting, updates the historical mean at each step, and follows the same walk-forward validation procedure as the tested models. A value < 1 indicates the model outperformed the benchmark; a value > 1 indicates the benchmark outperformed the model. Lower Relative MSE corresponds to a better rank.

This finding aligns with the broader forecasting literature, which often finds that simple models can be surprisingly effective for long-horizon forecasts of trending variables, while more sophisticated approaches add value primarily at shorter horizons where they can better capture cyclical patterns and temporary deviations.

4. Limitations

Several limitations impact the interpretation and application of these findings:

- **Data Latency and Timing:** The inherent latency in economic data releases means forecasts are always based on slightly outdated information. This project was conducted

during a time of significant economic volatility, with the most recent data point being from October 2024, making it challenging to fully incorporate the most recent market dynamics. As I mentioned in my presentation, "it's kind of a bad time to choose to do this project" given the current structural shifts in the economy.

- **Structural Breaks:** Post-COVID economic conditions present significant forecasting challenges that may not be fully captured by the models or the included dummy variables. We are potentially in the midst of another structural break right now, which could impact the accuracy of any forecast regardless of methodology.
- **Feature Selection Results:** The Lasso/ElasticNet results, heavily favoring lagged house prices, suggest that the linear relationships captured by these models didn't strongly incorporate other economic factors, although non-linear models might find different influences. Interestingly, even variables like unemployment rate at lag 4, which showed some significance in Granger causality tests, didn't appear as important in the feature selection process and including them actually worsened model performance.
- **VECM/VAR Implementation Challenges:** The implementation of VECM/VAR models faced significant hurdles due to diagnostic failures, preventing their inclusion in the final comparison. Working with these complex econometric models revealed that they require extremely rigorous understanding of their underlying assumptions and technical implementation details.

- **Hyperparameter Tuning:** While hyperparameter tuning was performed, it is computationally intensive, and more exhaustive searches could potentially yield further, likely marginal, improvements, always balancing against the risk of overfitting. In some cases, expanding the hyperparameter search space actually led to worse performance due to overfitting.
- **Alternative Transformations:** Time constraints precluded exploring log-differencing as a potential remedy for the VECM/VAR models, which might have yielded different results. Implementing this would have required redoing much of the analysis from scratch.

5. Future Work

Based on the findings and limitations, several avenues for future work emerge:

- **Alternative Transformations:** Re-evaluating models using log differences instead of percentage changes could potentially improve performance, especially for VECM/VAR models that were sensitive to the data's statistical properties.
- **Formal Structural Break Analysis:** Employing formal structural break tests could lead to more sophisticated modeling approaches like regime-switching models that explicitly account for different economic environments.
- **Advanced Feature Engineering:** Exploring interactions, non-linear transformations, and alternative machine learning models (e.g., Gradient Boosting) could uncover different

relationships not captured by the current models.

- **Ensemble Methods:** Combining forecasts from the best models like SVR and ElasticNet, potentially using different weights at different forecast horizons based on their relative strengths, might enhance robustness.
- **Improved Exogenous Variable Forecasting:** Developing more sophisticated approaches for projecting exogenous variables might improve multi-step forecasting performance.
- **LSTM Refinement:** Further architectural exploration and optimization of the LSTM model could improve its performance beyond the initial implementation.

6. What I Learned

This capstone project offered substantial learning opportunities and personal growth:

- **Data Handling Complexity:** I gained practical experience in the complexities of time series data management, including handling different frequencies and addressing missing data through various imputation strategies. I initially started with simple forward filling, but quickly realized the limitations of this approach, especially for economic variables.
- **Stationarity and Transformation:** Applying stationarity concepts and tests (ADF/KPSS) and understanding the impact of transformations like differencing and percentage changes was crucial. I learned how to handle these aspects both theoretically and

practically.

- **Robust Evaluation Techniques:** Implementing walk-forward validation and benchmarking highlighted their necessity for reliable time series assessment compared to standard cross-validation or simple metrics. As I mentioned in my presentation, "I can't believe I was gonna submit it two times a month ago using an R-squared curve without walk-forward validation."
- **Econometric Model Complexity:** Encountering the real-world challenges of applying complex econometric models (VECM/VAR) when underlying assumptions are violated was extremely instructive. I realized that economists who specialize in these techniques have incredibly deep expertise - as I put it in my presentation, "I realized economists that do this type of stuff are crazy big brain. Wow. They know a lot and it requires a very, very rigid understanding of certain things."
- **Machine Learning Effectiveness:** I gained appreciation for the effectiveness of sometimes simpler machine learning approaches (SVR) and the specific strengths of others (regularized regression for feature selection).
- **Hyperparameter Tuning Techniques:** The project exposed me to various hyperparameter tuning approaches including RandomizedSearchCV for Lasso/ElasticNet and Bayesian optimization for LSTM, along with understanding the critical balance between

optimization and overfitting avoidance.

- **Forecasting Non-Linearity:** I learned that forecasting is not a linear process - you don't just fit a model and evaluate R-squared. The iterative process of walk-forward validation is essential for any serious forecasting endeavor.
- **Value of Research Literature:** Consulting prior research proved invaluable for methodology. While AI tools were helpful for learning certain concepts, they couldn't replace the depth of domain-specific knowledge found in research papers like Milunovich's work on Australian housing markets.
- **Domain Knowledge Importance:** The project underscored how crucial domain expertise is for economic forecasting. Understanding the nuances of housing markets, broader economic indicators, and their interrelationships was essential for making informed modeling decisions.

7. Conclusion

This project successfully compared multiple forecasting models for the DC-area House Price Index. Through a rigorous walk-forward validation process benchmarked against a historical mean, a univariate Support Vector Regression (SVR) with 3 lags was identified as the most effective model for this specific dataset and forecast horizon. The final forecast generated using

SVR predicts continued, though potentially moderating, growth through Q4 2026, reaching approximately 424.88.

From a practical perspective, the strong autoregressive nature of the housing index where future values are primarily determined by recent past values rather than other economic indicators suggests that the market maintains its own momentum to a significant degree. For potential homebuyers in the DC area, this implies that the best time to buy may simply be when personal financial circumstances allow, as the market is projected to continue its upward trajectory.

The difficulties encountered when applying standard VECM/VAR models highlight how complex econometric approaches don't always translate well to real-world forecasting challenges, especially in the presence of structural breaks or other data complexities. Sometimes, more focused machine learning approaches that make fewer assumptions about the underlying data structure can produce superior results.

Overall, the project provided comprehensive practical experience in the end-to-end time series analysis and forecasting workflow, from data collection and preprocessing through model selection, validation, and interpretation. This experience has significantly deepened my understanding of both the technical aspects of forecasting and the practical challenges of applying these techniques to real-world economic data.

8. References

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